Evidence-Grounded Protocol Research Agent

Pancreatic / islet organoids: protocol assessment, phenotype analysis and testable optimization hypotheses

Domain results + evidence references return to Review

Registered action dispatch

A Research Inputs

Research question
Objects, endpoints, constraints

Structured protocols
Upstream-exported JSONL
Source quotes and identifiers

AnnData / h5ad
Expression + annotations
Sample / study / protocol links

Mechanism literature
Provided sources
Optional PubMed abstracts

Available execution modes

Chat: question-driven actions
Batch: PA > DA > PO

Extraction is upstream, not integrated.

Question

B Orchestrator

Single bounded Plan / Execute / Review loop

Configure
Identity, project scope, run snapshots

Plan LLM
Goal, objects, metrics, result bindings

Execute
Dispatch available registered actions

Review LLM
Evidence grounding + goal acceptance

Finalize
Source-bound conclusions + sealed report

Action Registry: availability / inputs / outputs

Bounded repair; local failures; clarification

C1 Protocol Analysis Agent (PA)

Source-grounded representation and comparison

Protocol Structure & Step Semantics Optional LLM
Schema, timeline, factor events;
scoped operations and parameters

Deterministic Feature Compiler
Feature specs, numeric values, provenance, audit

Outputs: matrix + original steps + source evidence

C2 Data Analysis Agent (DA)

Explicit tasks; deterministic tools — Optional LLM

Task Resolver & Phenotype Tools
QC; composition, expression, markers;
gene programs and differential expression

Maturity & Metric Validation
Early / Late scores; task-specific validity;
statistical applicability

Outputs: Phenotype IR + numeric tables + validation

D Protocol Optimizing Agent (PO)

Evidence-to-hypothesis synthesis

Joint Evidence Builder
Protocol identity / version + endpoint cohorts

Association & Sensitivity
Raw endpoints; replicates; study removal

Mechanism Evidence Integrator Optional LLM
Support / opposition;
alternative explanations

Hypothesis & Candidate Design
Intervention, control, endpoint, falsification

Critique & Conditional Selection
Sourced utility; missing assessments
remain unknown / unranked

Similar observed protocols are references,
not predictions of a new intervention.

Association is not causation.

Report

E Research Outputs

Finalized by Orchestrator

Protocol differences
Features + steps + context

Phenotype assessment
Endpoint-specific comparisons

Scientific explanations
Support, opposition, limitations

Testable hypotheses
Validation-ready specifications

Traceable artifacts
Versioned JSON, tables, reports

Evidence gaps remain explicit.

No guaranteed improvement.
No automatic wet-lab execution.

Read / persist

Shared Scientific State & Evidence Governance

Versioned Scientific Records

Protocol / Phenotype / Evidence / Claim / Hypothesis
Historical counterevidence; user / project isolation

Result Authority & Reproducibility

Hash-verified saved JSON + field-level references
Input, code, tool and model versions; immutable archives

Scientific & Execution Gates

Source coverage / QC / comparability / goal acceptance
Unknowns preserved; budget / audit / clarification

Optional external validation: supported extension, not an autonomous experimental loop

Researcher Review → External Culture / Experiment / Sequencing → QC-Passed Feedback Admission

LLM: planning / qualitative interpretation

Tools: numerical authority

Dashed: external, optional validation

Current-code framework | 2026-10-07 | Module names summarize responsibilities; they do not introduce extra agents.